

EXPERIMENT 1-QUICK SORT

CODE:-

```
#include <stdio.h>

void quickSort(int arr[], int low, int high) {
    if (low < high) {
        int pivot = arr[high];
        int i = low - 1;

        for (int j = low; j < high; j++) {
            if (arr[j] <= pivot) {
                i++;
                int temp = arr[i];
                arr[i] = arr[j];
                arr[j] = temp;
            }
        }
        int temp = arr[i + 1];
        arr[i + 1] = arr[high];
        arr[high] = temp;

        int pi = i + 1;

        quickSort(arr, low, pi - 1);
        quickSort(arr, pi + 1, high);
    }
}

// Function to print an array
void printArray(int arr[], int size) {
    for (int i = 0; i < size; i++) {
```

```
printf("%d ", arr[i]);  
    }  
printf("\n");  
}  
int main() {  
    int n;  
    // Input the size of the array  
    scanf("%d", &n);  
    int arr[n];  
    // Input the elements of the array  
    for (int i = 0; i < n; i++) {  
        scanf("%d", &arr[i]);  
    }  
    // Print the original array  
    printArray(arr, n);  
    // Sort the array using quick sort  
    quickSort(arr, 0, n - 1);  
    // Print the sorted array  
    printArray(arr, n);  
    return 0;  
}
```

SCREENSHOT :-TEST CASES

The screenshot shows a web browser window with the URL `sitnagpur.codetanttra.com/secure/course.jsp?euclid=6a323146ac24b7674600cfd#contents/6a323146ac24b7674600cfd/6a323146ac24b7674600cfd/6790b559d94dccc16...`. The page is titled "1.1.1. Quick Sort" and contains the following information:

Write a program to sort an array of length n using the Quick Sort algorithm. The program should read the array elements and print the array before and after sorting.

Input Format:

- The first line contains a single integer n , representing the number of elements in the array.
- The second line contains n space-separated integers representing the elements of the array.

Output Format:

- The first line of output contains space-separated integers representing the elements of the original array
- The Second input line contains space-separated integers representing the elements after sorting.

Constraints:

- $1 \leq n \leq 1000$

Note:

- The driver code is already provided. You need to complete the required code to perform Quick Sort correctly.

Sample Test Cases

The code editor shows the following C code:

```
1 #include <stdio.h>
2 // Quick Sort function
3 //void quickSort(int arr[], int low, int high) {
4 //}
5 void quickSort(int arr[], int low, int high) {
6     if (low < high) {
7         int pivot = arr[high];
8         int i = low - 1;
9
10        for (int j = low; j < high; j++) {
11            if (arr[j] <= pivot) {
```

The test cases section shows the following results:

Test case	Expected output	Actual output	Status
1	4 1 8 3 4	4 1 8 3 4	Passed
2	1 8 3 4 1 3 4 8	1 8 3 4 1 3 4 8	Passed

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